Mucahit Gemici

Ph.D. Candidate in Computer Science (Human–Computer Interaction), Concordia University, Montreal, Canada
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Education

Ph.D. in Computer Science, Concordia University, Montreal, Canada Jan 2024 – Dec 2026 (expected)

- Thesis: *One Workspace, Two Realities: Extending Desktop 3D Modeling with Augmented Reality and Artificial Intelligence*
- Thesis defence scheduled for December 7, 2026
- Advisor: Dr. Anil Ufuk Batmaz
- Research focus: hand-tracking failures in VR, immersive AI assistive systems, hybrid user interfaces

B.S. in Electrical & Electronics Engineering, Kadir Has University, Istanbul, Turkey 2022

Research Experience

Visiting Researcher, Nara Institute of Science and Technology (NAIST), Nara, Japan Jun 2026 – Sep 2026

- Worked in Prof. Kiyoshi Kiyokawa's Cybernetics and Reality Engineering (CARE) Laboratory, funded by the Mitacs Globalink Research Award.
- Developed an assistance system for daily use on lightweight smart glasses (paper in preparation for IEEE TVCG).

Ph.D. Researcher, Concordia University, Montreal, Canada Jan 2024 – present

- **Google-funded projects:** Engineered an early-warning system for common hand-tracking failures in VR, which detects tracking loss before it occurs and warns users with visual, auditory, or haptic feedback. Used eye tracking and user behavior logging to analyze reactions to the warnings, and studied user trust and confidence under false warnings. Resulted in five papers: three as first author (PLOS ONE; IEEE Access; under review at IEEE VR 2027) and two as co-author (IEEE TVCG, ISMAR 2026 Special Issue; under review at IEEE TVCG).
- **Mid-air object manipulation:** Designed and validated Object Speed Control with a signed distance field to improve precision in distant mid-air manipulation (IEEE ISMAR 2024).
- **Cross-reality 3D modeling:** Built MAR, a cross-reality 3D modeling system that pairs desktop Blender with AR inspection and gesture-based selection, and evaluated it with 20 participants (IEEE TVCG, ISMAR 2026 Special Issue).
- **Cross-reality selection:** Designed a mouse-guided snapping technique for selecting vertices, edges, and faces in AR, and evaluated it with 24 participants (in preparation for IJHCI).
- **AI-assisted 3D modeling:** Built a cross-reality modeling assistant that lets users refer to a part of a shape by describing, selecting, or drawing on it, and evaluated it with 24 participants (in preparation for IJHCI).
- **AI-assisted authoring in XR:** Designed an AR second-screen system for LLM-assisted desktop workflows and quantified the cost of crossing between physical and virtual workspaces (ACM SUI 2026).
- Developed experimental testbeds in Unity (C#) and conducted mixed-methods user studies for each project.

Undergraduate Researcher, Kadir Has University, Istanbul, Turkey 2021 – 2022

- Built MuscleNET, a wearable system that uses Arduino/MyoWare sensors with deep neural networks and LSTM networks to predict injuries and evaluate training quality.

Publications

First-authored

1. **Gemici, M.**, Johnsen, K., Barrera Machuca, M. D., & Batmaz, A. U. (2026). Modeling Across Realities (MAR): Extending Desktop 3D Modeling with Augmented Reality. *IEEE Transactions on Visualization and Computer Graphics* (ISMAR 2026 Special Issue), early access. doi:10.1109/TVCG.2026.3747484
2. **Gemici, M.**, Bashar, M. R., Stuerzlinger, W., Phadnis, V., & Batmaz, A. U. (2026). Crossing Realities Has a Cost: Evaluating AR Second Screens for AI-Assisted Desktop Authoring. In *Proceedings of the ACM Symposium on Spatial User Interaction (SUI '26)*. ACM. doi:10.1145/3822518.3830046
3. **Gemici, M.**, Hatira, A., Phadnis, V., & Batmaz, A. U. (2025). Gaze Analysis in Early Warning Visual Feedback System for Hand Tracking Failures in Virtual Reality. *IEEE Access, 13*. doi:10.1109/ACCESS.2025.3621645
4. **Gemici, M.**, Phadnis, V., & Batmaz, A. U. (2025). Before hands disappear: Effect of early warning visual feedback method for hand tracking failures in virtual reality. *PLOS ONE, 20*(6), e0323796. doi:10.1371/journal.pone.0323796

5. **Gemici, M.**, Stuerzlinger, W., & Batmaz, A. U. (2024). Object Speed Control With a Signed Distance Field for Distant Mid-Air Object Manipulation in Virtual Reality. In *2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)* (pp. 465–474). IEEE. doi:10.1109/ISMAR62088.2024.00061
6. **Gemici, M.**, Korkmaz, K., Ayhan, N. T., Soylu, S., Guc, F., & Ogrenci, A. S. (2022). MuscleNET: Smart Predictive Analysis for Muscular Activity Using Wearable Sensors. In *2022 Innovations in Intelligent Systems and Applications Conference (ASYU)*. IEEE. doi:10.1109/ASYU56188.2022.9925553

Co-authored

1. Hatira, A., Rasoulifar, Z., **Gemici, M.**, Phadnis, V., & Batmaz, A. U. (in press). Effects of false positive and false negative early warnings for hand tracking failures on user trust and confidence in virtual reality. *IEEE Transactions on Visualization and Computer Graphics* (ISMAR 2026 Special Issue). doi:10.1109/TVCG.2026.3747526
2. Doesburg, D., **Gemici, M.**, & Batmaz, A. U. (2025). Landing windows method as soft visual constraints for mid-air interactions. In *Proceedings of the 2025 ACM Symposium on Spatial User Interaction (SUI '25)* (pp. 1–12). ACM. doi:10.1145/3694907.3765922
3. Voisard, L., Hatira, A., Bashar, M. R., **Gemici, M.**, Sarac, M., Kersten-Oertel, M., & Batmaz, A. U. (2024). Subtask-based virtual hand visualization method for enhanced user accuracy in virtual reality environments. In *2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)* (pp. 6–11). IEEE. doi:10.1109/VRW62533.2024.00008

Under review

1. **Gemici, M.**, Phadnis, V., & Batmaz, A. U. Auditory early warnings for hand-tracking failures in VR [IEEE VR 2027].
2. Hatira, A., **Gemici, M.**, Phadnis, V., & Batmaz, A. U. Haptic early warnings for hand-tracking failures in VR [IEEE TVCG].

In preparation

1. **Gemici, M.**, Barrera Machuca, M. D., Johnsen, K., & Batmaz, A. U. Mouse-guided selection of mesh elements for cross-reality 3D modeling [Target: International Journal of Human–Computer Interaction (IJHCI)].
2. **Gemici, M.**, Doesburg, D., Johnsen, K., Barrera Machuca, M. D., & Batmaz, A. U. Prompting to edit 3D shapes in AI-assisted cross-reality modeling [Target: IJHCI].
3. **Gemici, M.**, Batmaz, A. U., & Kiyokawa, K. Proactive assistance on lightweight smart glasses [Target: IEEE TVCG].

Talks

- *Modeling Across Realities (MAR): Extending Desktop 3D Modeling with Augmented Reality.* IEEE ISMAR 2026 (TVCG journal paper), Bari, Italy · Oct 2026
- *Crossing Realities Has a Cost: Evaluating AR Second Screens for AI-Assisted Desktop Authoring.* ACM SUI 2026, Bari, Italy · Oct 2026
- *From Desktop to Smart Glasses: Designing Hybrid and AI-Assisted Interaction.* Research talk, Cybernetics and Reality Engineering (CARE) Laboratory, Nara Institute of Science and Technology (NAIST), Nara, Japan · Jun 2026

Awards & Honors

- **Conference and Exposition Allowance** (CAD $2,000), Concordia University · Aug 2026. Supported presenting at ACM SUI 2026 and IEEE ISMAR 2026.
- **Mitacs Globalink Research Award** (CAD $6,000), Mitacs · 2026. Competitive national award that funded the visiting researcher position at NAIST, Japan.
- **International Tuition Award of Excellence** (CAD $44,893), Concordia University · 2024. Tuition remission over eight terms that waives international tuition fees for the Ph.D.
- **Best Graduation Project Award**, Kadir Has University · 2022. Awarded for MuscleNET, a wearable system for predictive analysis of muscular activity (IEEE ASYU 2022).
- **Top Ranking Student**, Kadir Has University · 2022. Graduated first among all departments and faculties (GPA 3.94/4.00).

Teaching Experience

Teaching Assistant, Concordia University, Montreal, Canada Jan 2024 – present

Course	Term	Role
COMP 476: Advanced Game Development	Winter 2026	Lab Instructor
SOEN 287: Web Programming	Winter 2026	Marker
COMP 376: Introduction to Game Development	Fall 2025	Lab Instructor (2 sections) and Marker
COMP 476: Advanced Game Development	Winter 2025	Marker
COMP 376: Introduction to Game Development	Fall 2024	Lab Instructor
COMP 476: Advanced Game Development	Winter 2024	Lab Instructor (2 sections)

Service

Peer review (10 reviews)

- **ACM CHI:** Papers (2027), Posters (2026), Late-Breaking Work (2024, 2025)
- **Other conference papers:** IEEE VR 2027, IEEE ISMAR 2024, ACM VRST 2024, ACM MobileHCI 2024, ICXR 2026
- **Journals:** IEEE Transactions on Visualization and Computer Graphics (TVCG)

Volunteering

- **Student Volunteer Chair**, ACM Symposium on Virtual Reality Software and Technology (VRST '25) · Nov 2025. Coordinated the student volunteer team and managed conference merchandise (design, ordering, and shipping).
- **Student Volunteer**, ACM Symposium on Spatial User Interaction (SUI '26) · 2026
- **Event Volunteer**, GirlSET outreach program, Concordia University · 2024 and 2025. Presented research to the public to promote STEM engagement.

Professional Experience

Mobile Game Developer, Servo Studios, Istanbul, Turkey Jun 2022 – Jun 2023

- Engineered and shipped 5 mobile titles, reaching more than 10,000 downloads with a $0.31 cost per install.
- Optimized frame rate and memory use for low-end devices.
- Implemented telemetry systems to track user engagement.
- Collaborated with the publisher Moonee to prototype designs and UI/UX, turning wireframes into interactive applications.

Technical Skills

- **Interactive development:** Unity 3D (C#), Unreal Engine, XR SDKs (Meta, ARCore), Blender (Python scripting)
- **Hardware & IoT:** Arduino / ESP32, circuit design, soldering, signal processing
- **Research methods:** within-subjects experimental design and counterbalancing, power analysis (G*Power), repeated-measures and ART ANOVA, non-parametric tests (Friedman, Wilcoxon), standardized questionnaires (NASA-TLX, SUS, UEQ), thematic analysis, mixed-methods user studies
- **Data & tools:** Python, R, C / C++, LaTeX, Git, SPSS / JMP, Figma

References

Available upon request.